# Maurya Patel

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## EDUCATION

**Northeastern University(Boston): Master of Science in Computer Science [GPA: 3.88/4.0]** Jan 2023 - Dec 2024

**Dharmsinh Desai University: Bachelor of Technology in Computer Engineering [GPA: 8.48/10.0]** Jul 2018 - May 2022

## PROFESSIONAL EXPERIENCE

**Software Engineer at HubSpot (USA)** Feb 2026 –Oct 2026

- Built **LLM-powered media features** with **LiteLLM** and the **OpenAI Responses API** (prompt enhancement, title generation, fidelity inference) on **async workers**, and built **AI skills and tools** that expose image generation to HubSpot's Breeze assistant as an **agent-callable capability**.
- Built backend support for **context-aware AI image generation**, retrieving a user's blog or email content and screenshots as model context, and disabled generic prompt enhancement when context is present to avoid unrelated output.
- Served as DRI for **image-generation visual evals and model benchmarking**, comparing image models and validating changes with **Braintrust evals** and customer-satisfaction prompts to trace low-quality cases (such as logo requests) to prompt-template defaults.
- Built an **integration test suite** covering success, failure, and access-control flows that enabled **automatic production deploys** for an AI video service, replacing costly full end-to-end runs on every release.
- Led reliability work for **AI image generation workflows**: migrated status storage to **RocksDB** with 30-day TTL and added **read-through caching** that cut database load from frontend polling by 98%, de-risking rollout with branch deployments.
- Executed a **staged, zero-downtime migration of 91M clip suggestion records** to a normalized relational schema (dual-write, backfill, gated read cutover, legacy cleanup), validated across multiple production regions.

**Software Developer at State Street Corporation (USA)** Jan 2025 - Jan 2026

- Developed a **Python schema mapping service** with **FastAPI**: used **Pandas** to pre-process uploaded files, then passed the structured data to an **LLM** to map source columns to target schemas.
- Built a **Pandas-based transformation pipeline** supporting post-processing operations (**column renaming, filtering, custom queries**) with **versioned outputs**.
- Developed services for State Street's **middle-office data platform** on the **Data Quality** and **Data Reconciliation** teams, working within an **event-driven, Snowflake-based** data quality framework.
- Extended the Data Quality **event-consumer service** to publish **multiple events per consumed message** to **Cosmos DB**, and added a **parent-event ID** for end-to-end **lineage tracing**.
- Built an internal **service-to-service REST API** for **Data Quality report generation**, letting other services and workflows trigger reports programmatically.

**Software Developer Intern at State Street Corporation (USA)** Jan 2024 - Aug 2024

- Containerized the solution with **Docker** and integrated it into **Harness CI/CD** pipelines, with **Veracode** and **Black Duck** scans for secure deployment.
- Migrated the Data Quality and Reconciliation teams' pipelines from **Jenkins to Harness**, containerizing workflows with **Docker** and delivering **CI/CD pipelines** with clear documentation.
- Architected an **event-driven, microservices-based Email Service** to remove **single points of failure**, add real-time tracking, and improve reliability.

## PROJECTS

**Image Manipulation and Enhancement** | *(Skills: Java, Swing, JUnit, JavaFX, OOD, Git)* Jan 2023- April 2023

- Developed a **Spring Boot-based image processing application** with real-time histogram visualization and interactive GUI, leveraging **MVC and Command design patterns** for modularity and flexibility.
- Ensured **94% error resolution rate** through rigorous **JUnit testing**, improving code reliability and stability.

**Soundsage** | *(Skills: Python, Pandas, ML, EDA, React, Scikit-learn, Flask)* Sep 2024- Dec 2024

- Built a **personalized music recommendation system** using Spotify's 12M-song dataset with **KNN (Cosine Similarity, Euclidean Distance)** to suggest songs based on audio features.
- Developed a full stack**Flask-React interface** supporting playlist uploads, ratings, and real-time refinement of user profiles with lightweight ML models.

## TECHNICAL SKILLS

**Languages**: Python, Java, JavaScript, C++, Shell Scripting
**AI & LLM**: OpenAI Responses API, LiteLLM, LangChain, LangGraph, GEval, MLflow, scikit-learn
**Backend & Frameworks**: FastAPI, Spring Boot, Express, React, Streamlit, REST APIs, Event-Driven Microservices
**Data & Storage**: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, RocksDB, Elasticsearch, Snowflake, Apache Kafka
**Cloud & DevOps**: AWS, Azure, Docker, Kubernetes (AKS), Harness, Jenkins, Git, UNIX/Linux, FFmpeg
**Observability & Security**: Grafana, Prometheus, Veracode, Black Duck, CI/CD